

Phase 4 – Data Migration, Testing & Security:

Objective:

The primary objective of **Phase 4** is to automate the end-to-end backend lifecycle of network requests, enforce platform-level security access controls, and thoroughly validate data integrity and operational workflows.

This phase encompasses:

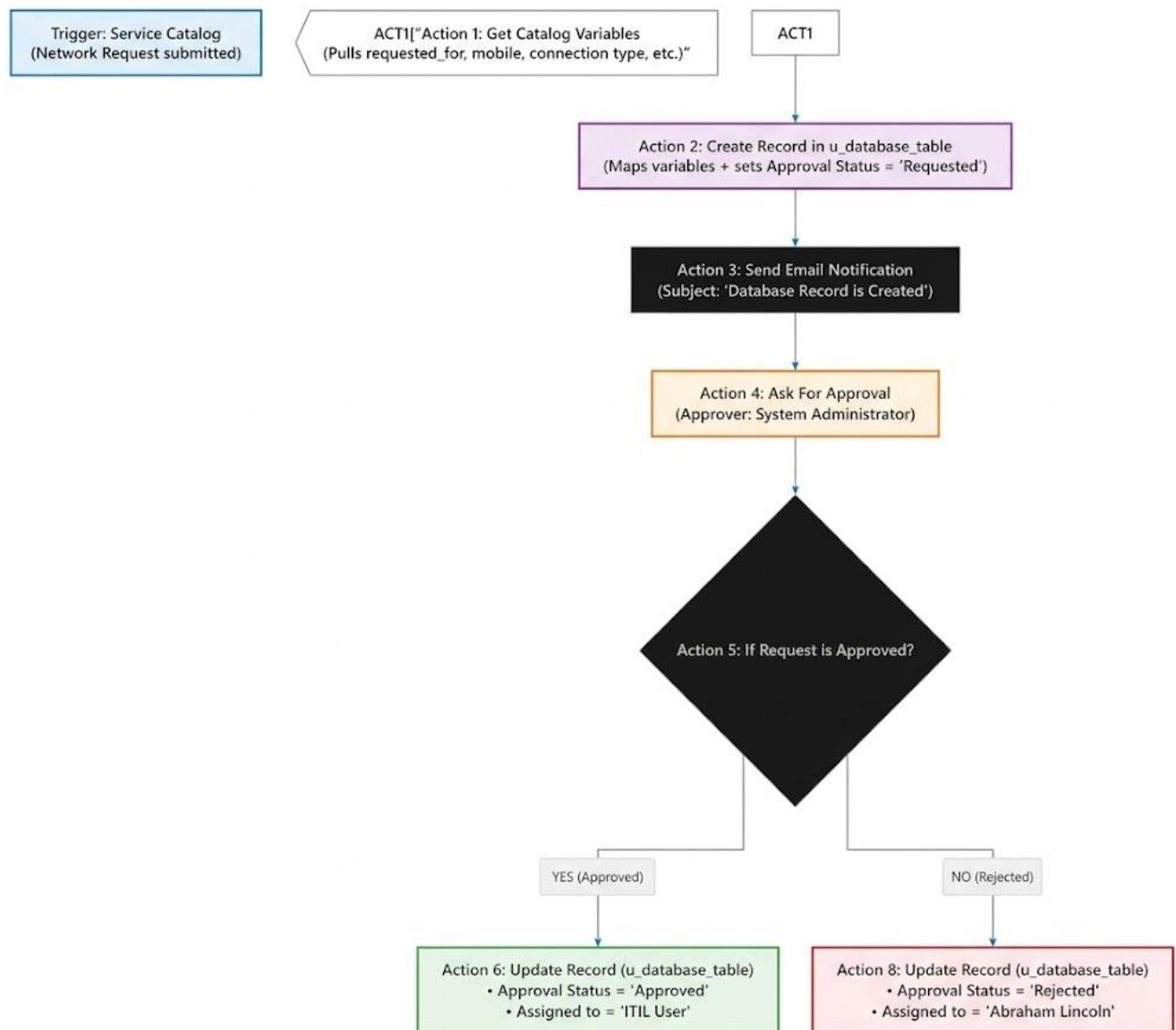
- **Workflow Automation:** Building an 8-step, zero-code automation flow in **Flow Designer** that captures catalog variables, synchronizes custom database records (u_database_table), triggers confirmation emails, routes approvals, and updates ticket states.
- **Security & Access Control (ACLs):** Verifying and enforcing table-level Access Control Lists to protect network records from unauthorized modifications or deletions.
- **Quality Assurance (QA) Testing:** Executing real-world lifecycle test simulations across positive (approval), negative (rejection), and dynamic UI test cases.
- **Data Integrity Safeguards:** Enforcing client- and server-side mandatory validations, automated dot-walk integrity, and conditional workflow state progression.

Automation Architecture (Flow Designer):

Flow Specifications:

- Flow Name: Request Flow
- Application Scope: Global
- Trigger: Service Catalog (executes automatically when the catalog item is submitted)
- Run As: System User
- Description: Automates record creation in u_request_management, sends email notifications, routes approvals, and updates assignment states.

Automation Architecture Flowchart



Step-by-Step Action Logic & Field Mappings:

Get Catalog Variables:

- Action: ServiceNow Core > Get Catalog Variables
- Submitted Request: Trigger - Service Catalog > Requested Item Record
- Template Catalog Item: Request

- Variables
Pulled: requested_for, mobile_number, type_of_connection, enter_your_existing_id, total_amount, mode_of_payment, address.

Create Record in Custom Table:

- Action: ServiceNow Core > Create Record
- Target Table: Request Management [u_request_management]
- Field Mapping Matrix:

Custom Table Field (u_database_table)	Source Data Pill / Value
Address (u_customer_address)	1 - Get Catalog Variables > address
Mobile number (u_mobile_number)	1 - Get Catalog Variables > mobile_number
Mode of Payment (u_mode_of_payment)	1 - Get Catalog Variables > mode_of_payment
Requested For (u_requested_for)	1 - Get Catalog Variables > requested_for
Total Amount (u_total_amount)	1 - Get Catalog Variables > total_amount
Type of Connection (u_type_of_connection)	1 - Get Catalog Variables > type_of_connection
Database Number (u_database_number)	Trigger > Requested Item Record > Number (e.g. REQ0010043)
Approval Status (u_approval_status)	Static String: Requested

Send Email Notification:

- **Action:** ServiceNow Core > Send Email
- **Target Record:** 2 - Create Record > Request Management Record
- **Include Watermark:** Checked (true)
- **To:** admin@example.com (or requester's email pill)
- **Subject:** Record is Created

Email Body Template:

Hello Yaswanth Busha,

We have received your request with request number: [Database Number].

Your request will be resolved within 2 business working days.

Thanks for contacting us,

Network Team.

Ask For Approval:

- **Action:** ServiceNow Core > Ask For Approval
- **Record:** 2 - Create Record > Request Management Record
- **Table:** Request Management [u_request_management]
- **Approval Reason:** Waiting for approval

- **Rules:** Approve when Anyone approves from System Administrator

Approval Path (IF Approved):

- **Flow Logic:** If: If Request is Approved
 - Condition: 4 - Ask For Approval > Approval State is Approved
- **(Inside then branch):** Update Record
 - Record: 2 - Create Record > Request Management Record
 - Assigned to: ITIL User (*Fulfillment engineer*)
 - Approval Status: Approved

Rejection Path (ELSE Rejected):

- **Flow Logic:** Else
- **Action 8 (Inside Else branch):** Update Record
 - Record: 2 - Create Record > Request Management Record
 - Assigned to: Yaswanth Busha (*Requester / Admin*)
 - Approval Status: Rejected

Security & Access Control (ACLs):

Table-Level Access Control Rules:

ACL Name	Operation	Applied Role	Description
u_request_management	Read	u_request_management_user, itil, admin	Allows requesters and support staff to query and view connection records.
u_request_management	Create	u_request_management_user, admin	Allows authenticated catalog users and system automation to create records.
u_request_management	Write	itil, admin	Restricts record updates (state changes, notes) exclusively to fulfillment teams.
u_request_management	Delete	Admin	Restricted strictly to System Administrators to prevent accidental data loss.

Role-Based Governance:

- Requesters (snc_internal / u_request_management_user): Can only submit requests via the portal and view their own submitted tickets. They cannot edit approval states or assignment fields.
- Fulfillment Engineers (itil): Can update fulfillment fields and complete implementation tasks once a ticket is approved.
- Approvers (admin / managers): Can render approval decisions via the Approval Request related list.

Testing & Request Lifecycle Verification:

Test Case ID	Test Scenario	Execution Steps & Input Data	Expected Result	Actual Outcome	Status
1	End-to-End Happy Path	1. Submit request via portal 2. Fill all fields (New connection) 3. Submit request	REQ generated; record created in u_request_management; email dispatched; approval pending.	Record created with status Requested; email logged in sys_email.	PASS
2	Approval Execution	1. Open record in u_request_management 2. Open Approval Request tab 3. Admin clicks Approve	Flow detects approval; status flips to Approved; Assigned to updates to ITIL User.	Status updated to Approved; Assigned to set to ITIL User.	PASS
3	Rejection Handling	1. Submit new request 2. Admin clicks Reject in related list	Status flips to Rejected; Assigned to set to requester; no fulfillment task assigned.	Status updated to Rejected; assigned to requester.	PASS
4	Outbound Email Verification	Check System Logs > Emails after submission	Confirmation email queued with watermark and request number.	Email sent with subject Database Record is Created.	PASS
5	Flow Execution Tracing	Inspect Flow Designer > Execution Details	Flow completes all steps with state Complete and no errors.	All green execution checkmarks.	PASS

Data Integrity Safeguards:

1. Client-Side & Server-Side Validations:

- Critical fields (mobile_number, total_amount, address, type_of_connection, mode_of_payment) are set to Mandatory, guaranteeing that empty payloads cannot enter the database.

2. Auto-Populate Data Integrity:

- Requester information is dot-walked directly from the active session (sys_user), completely eliminating spelling mistakes and invalid email addresses.

3. Approval State Enforcement:

- Flow Designer condition gates guarantee that no ticket can be marked Approved or assigned to ITIL User without a formal signature in sysapproval_approver.

4. Relational Synchronization:

- Automated mapping bridges the catalog request (sc_req_item) to the custom table (u_request_management), ensuring data consistency between ITSM requests and network operational registries.